

BMEG 3105 Fall 2026

Data analytics for personalized genomics and precision medicine

Scribe of lecture 3: The foundation of modern biology and genomics: Sequence data

1. Sequence data

- **Central dogma:** DNA → RNA → protein.
 - DNA replication: DNA → DNA
 - Transcription: DNA → RNA
 - Translation: RNA → protein
- Genetic information is stored in DNA sequences.
- **Phenotype = Genotype + Environment;** genotype is determined by sequences.
- **DNA:** A, T, C, G; double-stranded helix; ~3 billion base pairs in human genome.
- **RNA:** A, U, C, G.
- **Protein:** usually 20 amino acids; multiple sequence alignment.

2. How Sequences Are Obtained

- DNA/RNA sequencing: still developing; short reads → long reads.
- Nanopore sequencing:
 - DNA passes through a protein pore.
 - Different bases cause characteristic electrical current changes.
 - Very long reads (up to 4 Mb vs ~1000 bp).
 - Higher error rate (~0.25% vs ~0.001%).
- Protein sequencing:
 - Mostly mass spectrometry (MS).

- Break protein into short pieces → determine by MS → assemble like a jigsaw puzzle.

3. Sequence comparison and alignment score

- **Sequence alignment:** To determine the similarity between sequences and identify regions of similarity
 - Similar sequence → Similar structure → Similar function: Biomolecular function and property prediction
 - Similar sequence → Common ancestor: evolution, identifying conservative region, investigating mechanisms
- **Pairwise sequence alignment:** arrange two sequences to maximize similarity; gaps may be inserted.
- **Operations:**
 - Match: $A \leftrightarrow A$
 - Mismatch/substitution: $G \leftrightarrow T$
 - Gap/indel: $C \leftrightarrow _$
- Scoring matrix example:

	A	C	G	T
A	2	-7	-5	-7
C	-7	2	-7	-5
G	-5	-7	2	-7
T	-7	-5	-7	2

- Gap penalty = -10.
- Alignment score = sum of pair scores.
- Example: score = $2 + (-7) + 2 + 2 + (-10) + 2 = -9$.

4. Why dynamic programming?

- Straightforward enumeration of all alignments is impossible.
 - For 2 sequences of length n , the number of possible alignments is

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}$$
 - Dynamic programming greatly reduces computation time
- DP principles:
 - Break problem into smaller sub-problems.

- Solve sub-problems optimally and recursively.
- Use optimal sub-solutions to construct the optimal original solution.

5. Sequence Alignment with DP + DP Matrix

- Recurrence for prefixes $x[1..i]$, $y[1..j]$:
 - $$F(i, j) = \text{Max} \begin{cases} F(i-1, j-1) + S(x_i, y_j) \\ F(i-1, j) + gp \\ F(i, j-1) + gp \end{cases}$$
 - Boundary cases:
 1. $F(0, 0) = 0$
 2. $F(i, 0) = i \times gp$
 3. $F(0, j) = j \times gp$
- In the DP table, each cell stores: best score + arrow (path information)
- If there are ties, a cell may have multiple arrows, leading to multiple optimal alignments.
- Traceback:
 - Start from the last cell.
 - Follow arrows back to the origin.
 - ➔ Reconstruct the optimal alignment.
- Without arrows, we only know the optimal score, not the alignment itself.
- DP table thus stores both:
 - Sub-problem answers (scores).
 - Construction path (arrows).